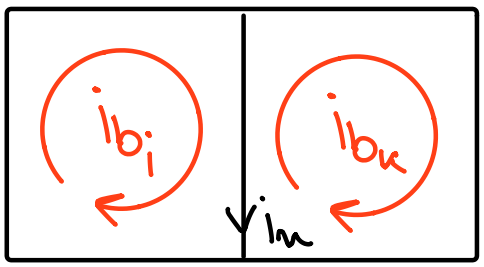
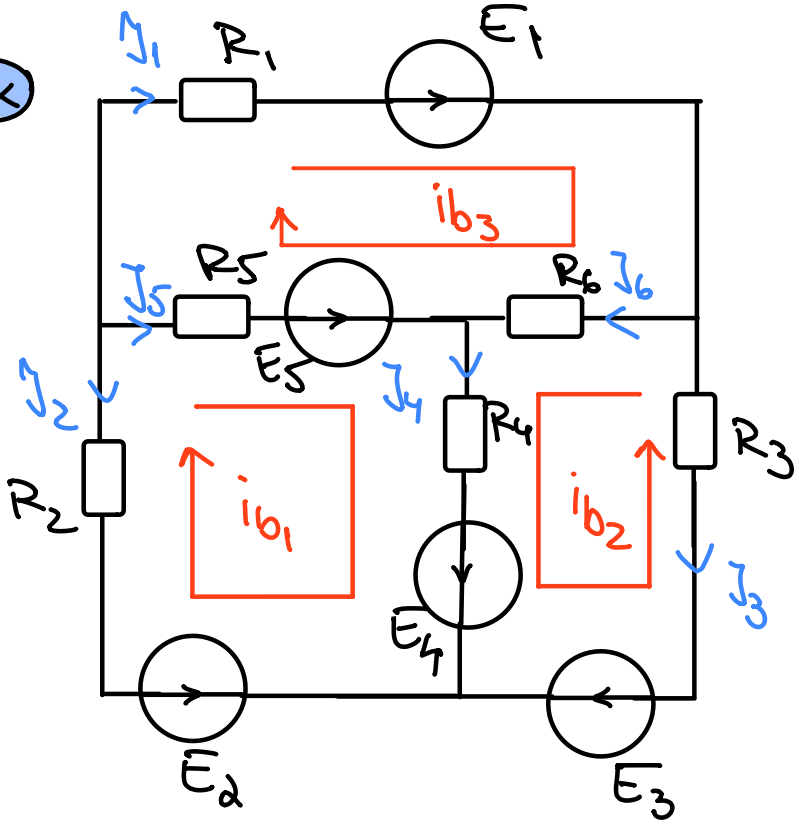
Metode de rezolvare a problemelor

I Metoda curenților ciclici (de buclă)



$$i_u = i_{b_j} - i_{b_k}$$

ex



$E_1 = 48\ V$
$E_2 = 76\ V$
$E_3 = 35\ V$
$E_4 = 6\ V$
$E_5 = 18\ V$
$R_1 = R_5 = 2\ \Omega$
$R_2 = 5\ \Omega$
$R_3 = R_4 = 3\ \Omega$
$R_6 = 4\ \Omega$

TK2: (V1)

$$(b_1): J_5R_5 - E_5 + J_4R_4 - E_4 + E_2 - J_2R_2 = 0$$
$$(b_2): J_6R_6 + J_4R_4 - E_4 + E_3 - J_3R_3 = 0$$
$$(b_3): J_1R_1 - E_1 + J_6R_6 + E_5 - J_5R_5 = 0$$

(V2)

$$\begin{cases} R_5(J_{b_1} - J_{b_3}) - E_5 + R_4(J_{b_1} + J_{b_2}) - E_4 + E_2 - R_2(-J_{b_1}) = 0 \\ R_6(J_{b_2} + J_{b_3}) + R_4(J_{b_1} + J_{b_2}) - E_4 + E_3 - R_3(-J_{b_2}) = 0 \\ R_1J_{b_3} - E_1 + R_6(J_{b_2} + J_{b_3}) + E_5 - R_5(J_{b_1} - J_{b_3}) = 0 \end{cases}$$

(V3)

$$\begin{cases} R_{11} I_{b1} + R_{12} I_{b2} + R_{13} I_{b3} = E_{b1} \\ R_{21} I_{b1} + R_{22} I_{b2} + R_{23} I_{b3} = E_{b2} \\ R_{31} I_{b1} + R_{32} I_{b2} + R_{33} I_{b3} = E_{b3} \end{cases}$$

$R_{11}, R_{22}, \dots R_{jj} = \sum_k R_k \quad k \in 1,2 \dots j$

(suma rezistențelor din bucla k)

$R_{jk} = R_{kj} = \pm \sum_i R_i \quad i \in (k \cup j)$

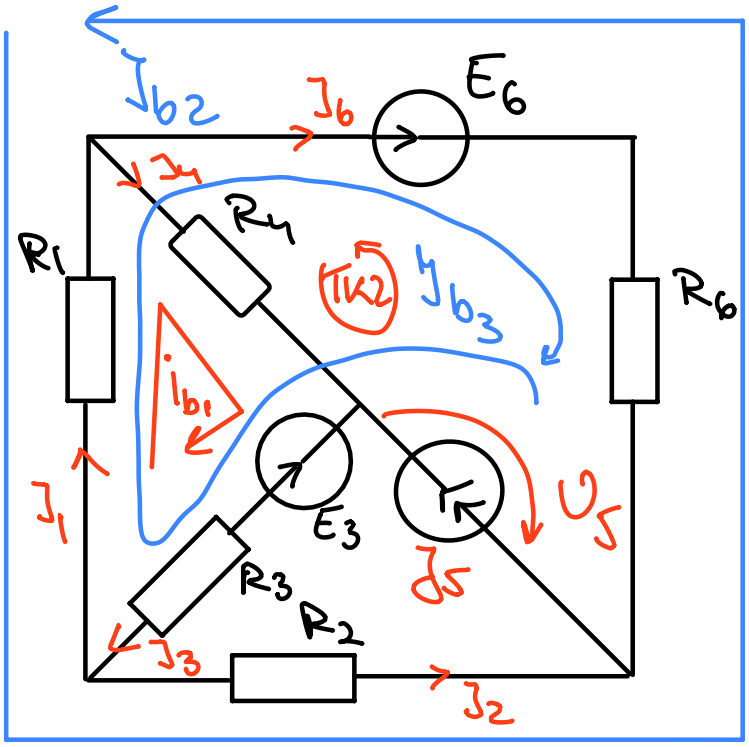
(suma rezistențelor comune buclelor k și j)

$$\begin{cases} I_{b1}(R_2 + R_4 + R_5) + I_{b2} R_4 + I_{b3}(-R_5) = E_5 + E_4 - E_2 \\ I_{b1} R_4 + I_{b2}(R_3 + R_4 + R_6) + I_{b3} R_6 = E_4 - E_3 \\ I_{b1}(-R_5) + I_{b2} R_6 + I_{b3}(R_1 + R_5 + R_6) = E_1 - E_5 \end{cases}$$

Obs

În cazul în care circuitul conține laturi cu surse de curent, se aleg curenții de buclă astfel încât prin latura cu sursa de curent să treacă un singur curent de buclă, valoarea acesteia fiind dată de valoarea sursei de curent.

ex



$R_1 = 15\,\Omega$

$R_2 = 6\,\Omega$

$R_3 = 5\,\Omega$

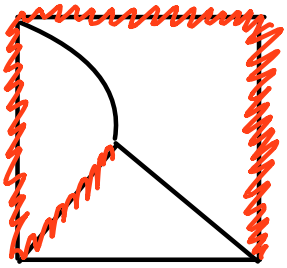
$E_3 = 20\,V$

$R_6 = 2\,\Omega$

$E_6 = 48\,V$

$J_5 = 1\,A$

$R_4 = 10\,\Omega$



V1 $(b_1)\ R_1 J_1 + R_4 J_4 - E_3 + R_3 J_3 = 0$

V2 $R_1(J_{b1} - J_{b2} + J_{b3}) + R_4 J_{b1} - E_3 + R_3(J_{b1} + J_{b3}) = 0$

$(b_2)\ -R_1 J_1 + R_2 J_2 - R_6 J_6 + E_6 = 0$

$-R_1(J_{b1} - J_{b2} + J_{b3}) + R_2 J_{b2} - R_6(J_{b3} - J_{b2}) + E_6 = 0$

$J_{b3} = J_5 = 1\,A$

$J_{b3} = J_5 = 1\,A$

V3
$$\begin{cases} J_{b1}(R_1 + R_3 + R_4) + J_{b2}(-R_1) + J_{b3}(R_1 + R_3) = E_3 \\ J_{b1}(-R_1) + J_{b2}(R_1 + R_2 + R_6) + J_{b3}(-R_1 - R_6) = -E_6 \end{cases}$$

$J_{b3} = 1\,A$

V3

$$\begin{cases} 30 \cdot J_{b1} - 15 J_{b2} + 1 \cdot 20 = 20 \rightarrow 2 J_{b1} = J_{b2} \\ -15 J_{b1} + 23 J_{b2} + 1 \cdot (-17) = -48 \end{cases} \Rightarrow$$

$$\Rightarrow -15 J_{b1} + 46 J_{b1} - 17 = -48 \Rightarrow 31 J_{b1} = -31$$

$$\Rightarrow J_{b1} = -1\,A$$

$$J_{b2} = -2\,A$$

$$J_1 = J_{b1} + J_{b3} - J_{b2} = -1 + 1 - 2 = -2\,A$$

$$J_2 = J_{b2} = -2\,A$$

$$J_3 = J_{b1} + J_{b3} = -1 + 1 = 0\,A$$

$$J_4 = J_{b1} = -1\,A$$

$$J_6 = J_{b3} - J_{b2} = 1 - (-2) = 3\,A$$

$U_5 = ?$ $\quad R_4 J_4 + U_5 - R_6 J_6 + E_6 = 0$

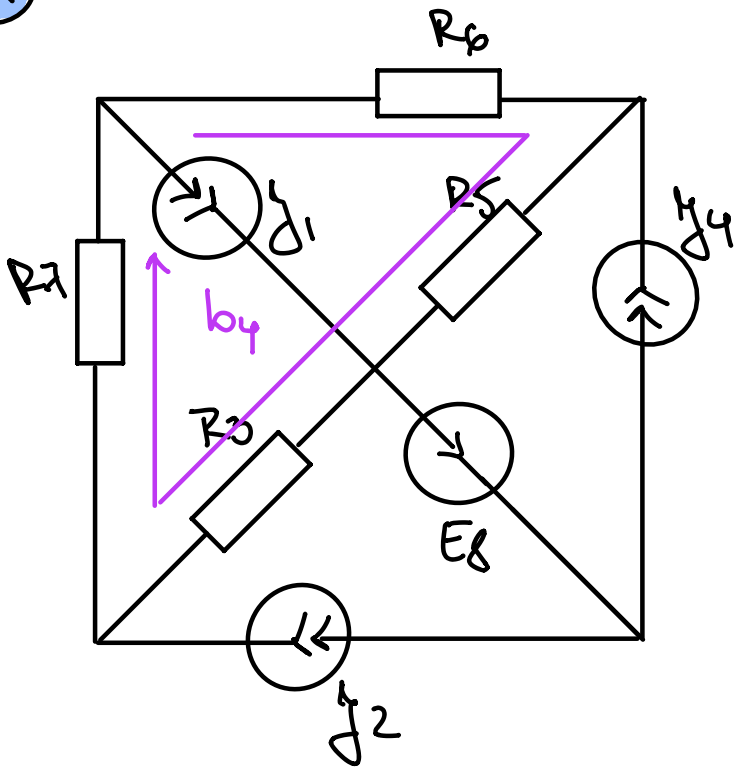
$$U_5 = 10 + 6 - 48 \Rightarrow U_5 = -32\,V$$

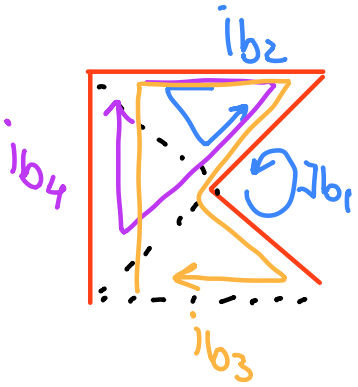
$$P_{gen} = E_6 J_6 + E_3 J_3 + U_5 J_5 = 48 \cdot 3 + 20 \cdot 0 + (-32) \cdot 1 = 112\,W$$

$$P_{cons} = R_1 J_1^2 + R_2 J_2^2 + R_3 J_3^2 + R_4 J_4^2 + R_6 J_6^2 =$$

$$= 15 \cdot 4 + 6 \cdot 4 + 5 \cdot 0 + 10 \cdot 1 + 2 \cdot 9 = 112\,W$$

ex

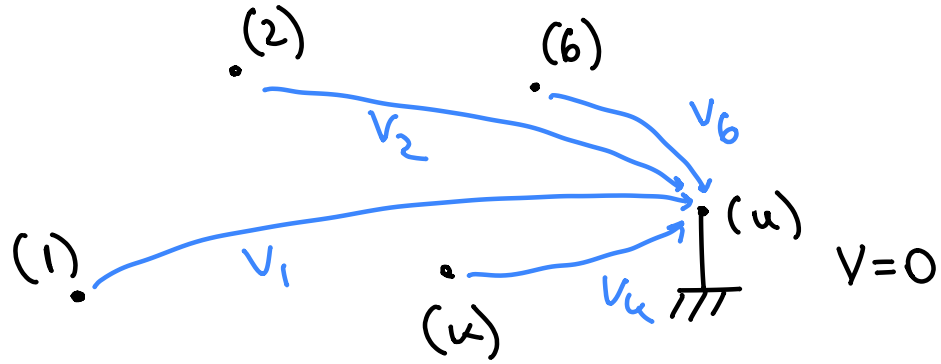




$$J_{b1} = J_4 \qquad J_{b2} = J_1 \qquad J_{b3} = J_2$$

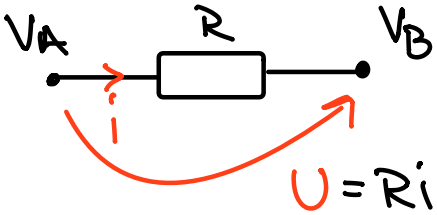
$$(b_4): R_7(J_{b3} + J_{b4}) + R_6(-J_{b2} + J_{b3} + J_{b4}) +$$
$$+ R_5(J_{b1} - J_{b2} + J_{b3} + J_{b4}) + R_3 J_{b4} = 0$$

II Metoda potențialelor nodurilor (Met. nodurilor)



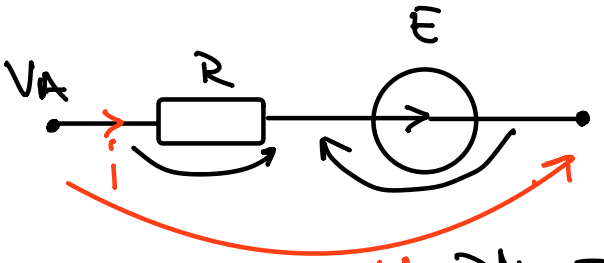


$$U = V_A - V_B$$



$$U = Ri \Rightarrow i = \frac{U}{R} \Rightarrow i = \frac{V_A - V_B}{R}$$

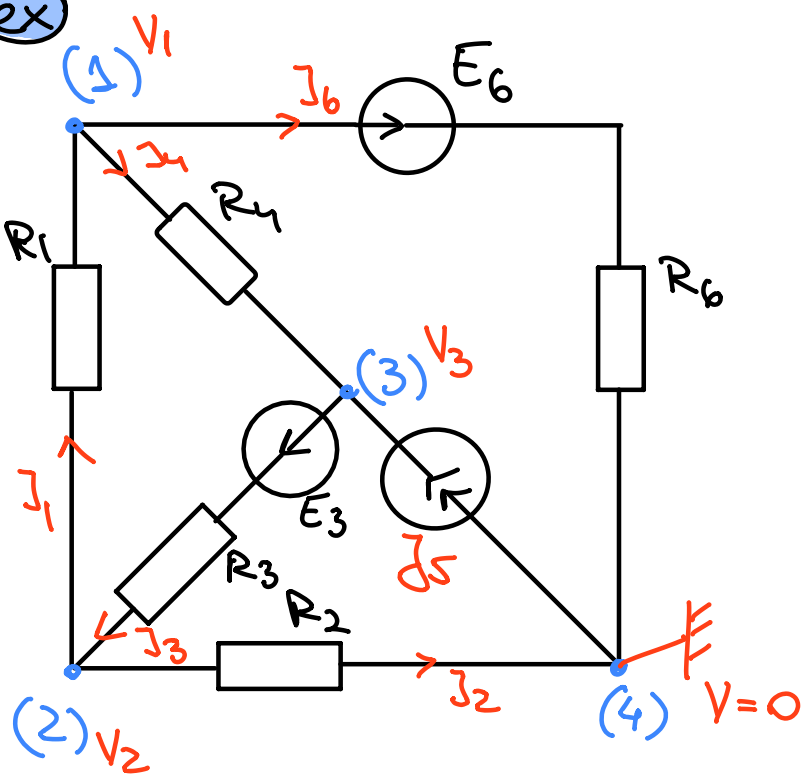
$$\boxed{i = \frac{V_A - V_B}{R}}$$



$$U = Ri - E \Rightarrow i = \frac{U+E}{R} \Rightarrow$$

$$\boxed{i = \frac{V_A - V_B + E}{R}}$$

ex



$R_1 = 15\Omega$

$R_2 = 6\Omega$

$R_3 = 5\Omega$

$E_3 = 20V$

$R_6 = 2\Omega$

$E_6 = 48\,V$

$J_5 = 1A$

$R_4 = 10\Omega$

V1

$$(1):\ I_6 + I_4 - I_1 = 0 \Rightarrow \frac{V_1 - 0 + E_6}{R_6} + \frac{V_1 - V_3}{R_4} - \frac{V_2 - V_1}{R_1} = 0$$

$$(2):\ I_1 + I_2 - I_3 = 0 \Rightarrow \frac{V_2 - V_1}{R_1} + \frac{V_2 - 0}{R_2} - \frac{V_3 - V_2 + E}{R_3} = 0$$

$$(3):\ I_3 - I_4 - J_5 = 0 \Rightarrow -\frac{V_1 - V_3}{R_3} + \frac{V_3 - V_2 + E_3}{R_3} - J_5 = 0$$

G = conductanța $\quad G = \frac{1}{R}$

I_{sc} = curent de scurtcircuit

V2

$$\begin{cases} G_{11}V_1 + G_{12}V_2 + G_{13}V_3 = I_{sc1} \\ G_{21}V_1 + G_{22}V_2 + G_{23}V_3 = I_{sc2} \\ G_{31}V_1 + G_{32}V_2 + G_{33}V_3 = I_{sc3} \end{cases}$$

$G_{11}, G_{22} \dots G_{jj} = \sum_k \frac{1}{R_k}$

(suma conductanțelor $\frac{1}{R}$ din nodul j)

$G_{jk} = G_{kj} = -\sum_i \frac{1}{R_i}$ $\quad i \in j \cap k$

(suma conductanțelor $\frac{1}{R}$ comune nodurilor $j \neq k$)

$$I_{sc_k} = \pm\left(\frac{E_k}{R_k} + J_k\right)$$

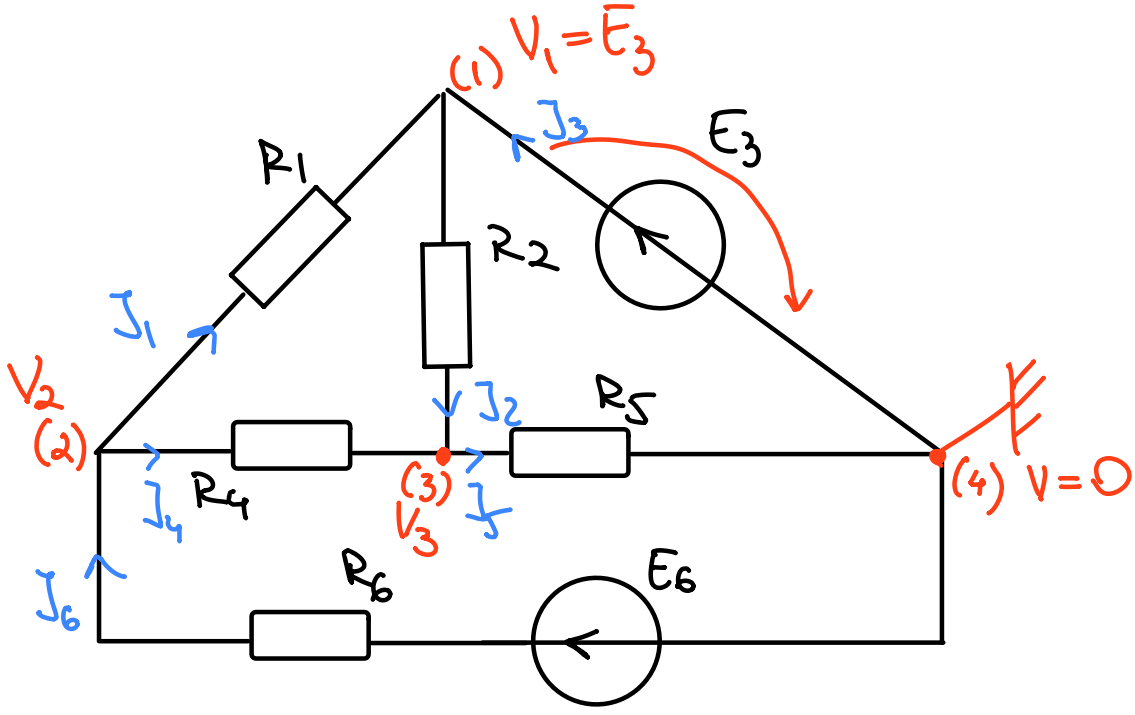
$$\begin{cases} V_1\cdot\left(\frac{1}{R_1}+\frac{1}{R_4}+\frac{1}{R_4}\right)+V_2\cdot\left(-\frac{1}{R_1}\right)+V_3\left(-\frac{1}{R_4}\right)=\frac{-E_6}{R_6} \\ V_1\left(-\frac{1}{R_1}\right)+V_2\left(\frac{1}{R_1}+\frac{1}{R_3}+\frac{1}{R_2}\right)+V_3\cdot\left(-\frac{1}{R_3}\right)=\frac{E_3}{R_3} \\ V_1\left(-\frac{1}{R_4}\right)+V_2\left(-\frac{1}{R_3}\right)+V_3\cdot\left(\frac{1}{R_3}+\frac{1}{R_4}+\frac{1}{\infty}\right)=J_J-\frac{E_3}{R_3} \end{cases}$$

Obs

- În cazul în care circuitul conține o latură doar cu sursă ideală de tens. ($R=0$) se alege ca nod de referință unul dintre nodurile laturii respective, potențialul calculat nod fiind dat de valoarea sursei

ex

$R_1=10\,\Omega$
$R_2=5\,\Omega$
$E_3=30\,V$
$R_4=10\,\Omega$
$R_5=5\,\Omega$
$R_6=2\,\Omega$
$E_6=46\,V$



V_1

(2) $I_1 + I_4 - I_6 = 0 \Rightarrow \frac{V_2 - V_1}{R_1} + \frac{V_2 - V_3}{R_4} - \frac{0 - V_2 + E_6}{R_6} = 0$

(3) $-I_2 - I_4 + I_5 = 0 \Rightarrow -\frac{V_1 - V_3}{R_2} - \frac{V_2 - V_3}{R_4} + \frac{V_3 - 0}{R_5} = 0$

$V_1 = E_3 = 30\ V$

$$\begin{cases} \frac{V_2 - 30}{10} + \frac{V_2 - V_3}{10} - \frac{-V_2 + 46}{2} = 0 \\ -\frac{30 - V_3}{5} - \frac{V_2 - V_3}{10} + \frac{V_3}{5} = 0 \end{cases}$$

$$\begin{cases} V_2 - 30 + V_2 - V_3 + 5V_2 - 5 \cdot 46 = 0 \\ -60 + 2V_3 - V_2 + V_3 + 2V_3 = 0 \end{cases}$$

$$\begin{cases} 7V_2 - V_3 = 30 + 230 \mid \cdot 5 \\ -V_2 + 5V_3 = 60 \end{cases} \Rightarrow \begin{cases} 35V_2 - 5V_3 = 1300 \\ -V_2 + 5V_3 = 60 \end{cases}$$

$$34V_2 = 1360 \Rightarrow V_2 = \frac{1360}{34} = 40\ V$$

$$V_3 = 7V_2 - 260 = 280 - 260 = 20\ V$$

V_2

$$\begin{cases} (1)\ V_1 = E_3 \\ (2)\ V_1\left(-\frac{1}{R_1}\right) + V_2\left(\frac{1}{R_1} + \frac{1}{R_4} + \frac{1}{R_6}\right) + V_3\left(-\frac{1}{R_4}\right) = \frac{E_6}{R_6} \\ (3)\ V_1\left(-\frac{1}{R_2}\right) + V_2\left(-\frac{1}{R_4}\right) + V_3\left(\frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{R_5}\right) = 0 \end{cases}$$

$$\begin{cases} V_1 = 30V \\ 30\left(-\frac{1}{10}\right) + V_2\left(\frac{1}{10}+\frac{1}{10}+\frac{1}{2}\right) + V_3\left(-\frac{1}{10}\right) = \frac{46}{2} \quad |\cdot 10 \\ 30\left(-\frac{1}{5}\right) + V_2\left(-\frac{1}{10}\right) + V_3\left(\frac{1}{5}+\frac{1}{10}+\frac{1}{5}\right) = 0 \quad |\cdot 10 \end{cases}$$

$$\begin{cases} V_1 = 30V \\ -30 + V_2(1+1+5) - V_3 = 46 \cdot 5 \\ 30(-2) - V_2 + V_3(2+1+2) = 0 \end{cases} \Rightarrow \begin{cases} 7V_2 - V_3 = 260 \quad |\cdot 5 \\ 5V_3 - V_2 = 60 \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} 35V_2 - 5V_3 = 260 \cdot 5 \\ -V_2 + 5V_3 = 60 \end{cases} \Rightarrow 34V_2 = 1300 + 60$$

$$V_2 = \frac{1360}{34} = 40V$$

$$V_3 = 7V_2 - 260 = 280 - 260 \Rightarrow V_3 = 20V$$

$$I_1 = \frac{V_2 - V_1}{R_1} = \frac{40-30}{10} = 1A$$

$$I_2 = \frac{V_1 - V_3}{R_2} = \frac{30-20}{5} = 2A$$

$$I_1 + I_3 = I_2 \Rightarrow I_3 = I_2 - I_1 = 2 - 1 = 1A$$

$$I_4 = \frac{V_2 - V_3}{R_4} = \frac{40-20}{10} = 2A$$

$$I_5 = I_2 + I_4 = 2 + 2 = 4A$$

$$I_6 = I_5 - I_3 = 4 - 1 = 3A$$